

Industrial Internship Report on " AI-Based Student Performance Prediction System"

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Executive Summary

This report provides details about the Industrial Internship conducted by upSkill Campus in collaboration with UniConverge Technologies Pvt Ltd (UCT). During this internship, a real-world project titled "AI-Based Student Performance Prediction System" was developed using Python and Machine Learning.

The project focuses on predicting student academic performance using factors such as study hours, attendance percentage, assignments completed, and previous marks. A Linear Regression Machine Learning model was used to analyze the data and generate predictions.

This internship provided practical exposure to Artificial Intelligence, Data Science, and Machine Learning concepts. It also improved problem-solving, programming, and project development skills.

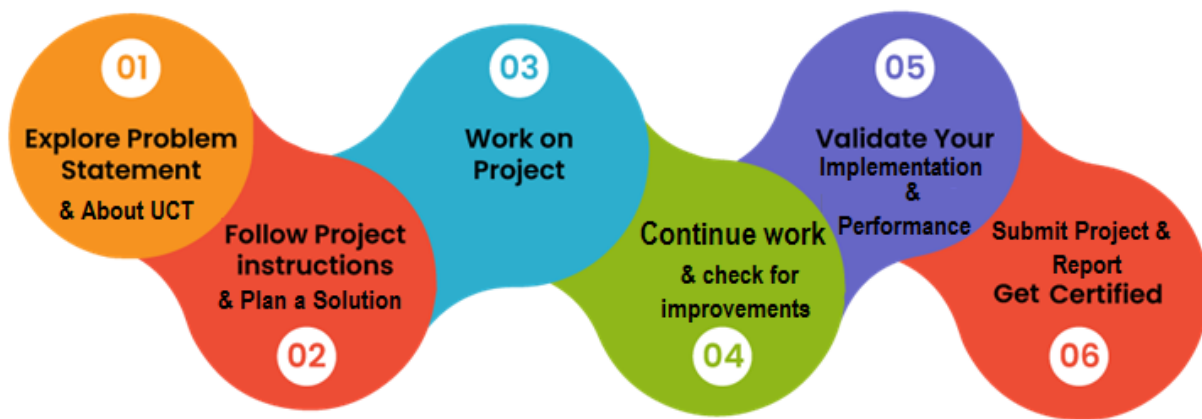
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1 Preface

The internship program helped in understanding real industrial project development and implementation. During the six weeks of internship, knowledge about Python programming, Machine Learning, data analysis, and project documentation was gained.

The project involved collecting student-related data, preprocessing the data, training a machine learning model, and predicting academic performance. The internship also improved communication skills, teamwork, and technical knowledge.



Special thanks to upSkill Campus and UniConverge Technologies Pvt Ltd for providing this opportunity and guidance throughout the internship.

2 Introduction

Artificial Intelligence and Machine Learning are transforming the education sector by helping institutions analyze student performance effectively. Predicting student results in advance helps teachers identify weak students and provide additional support.

The AI-Based Student Performance Prediction System uses machine learning algorithms to analyze student data and predict their final academic performance. This project demonstrates the practical application of data science in educational systems.

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



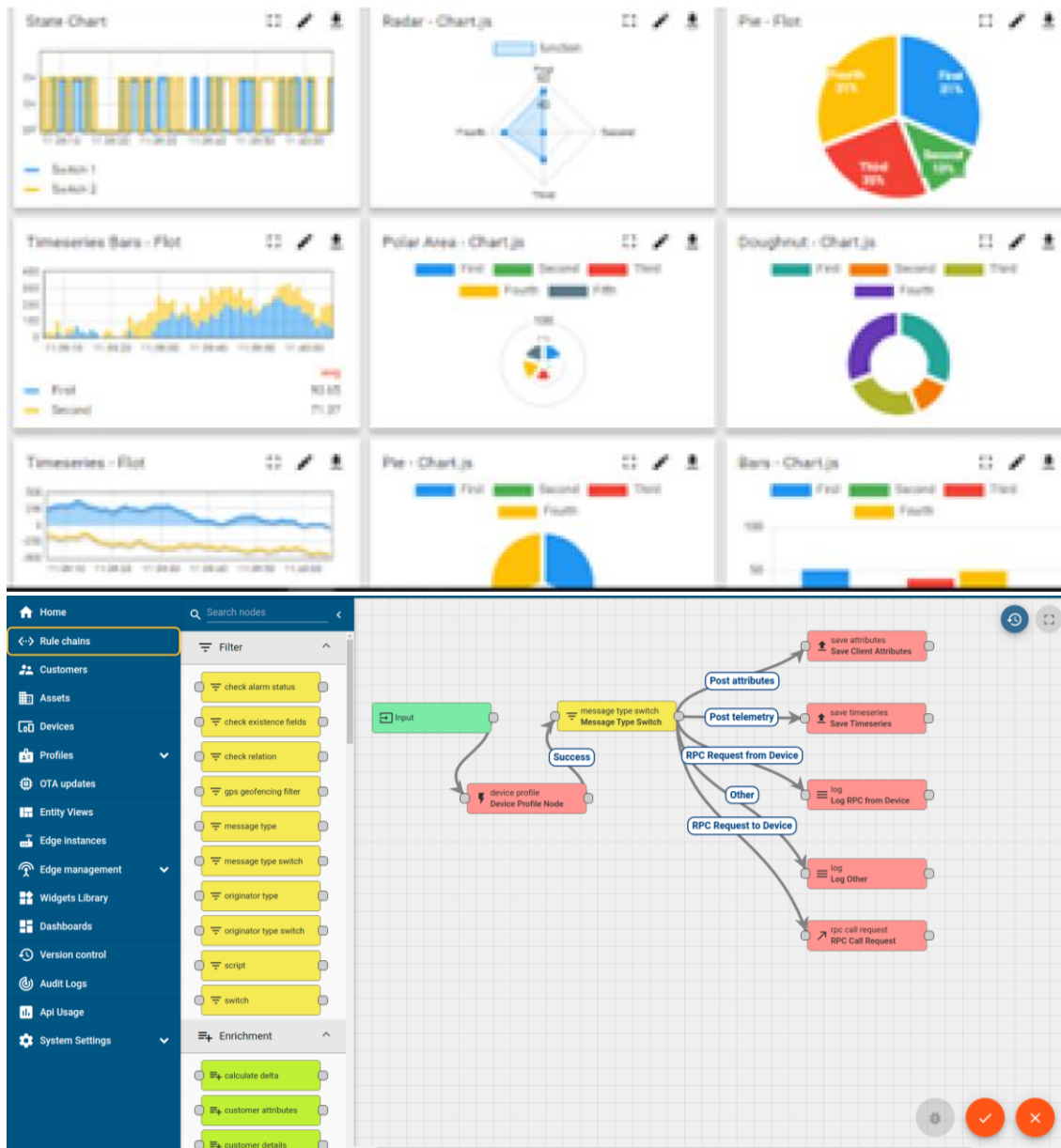
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



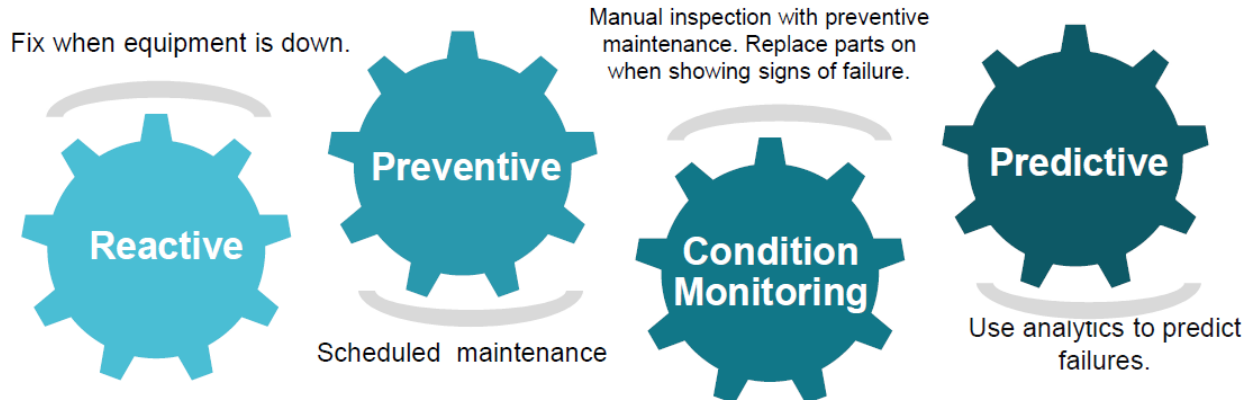


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

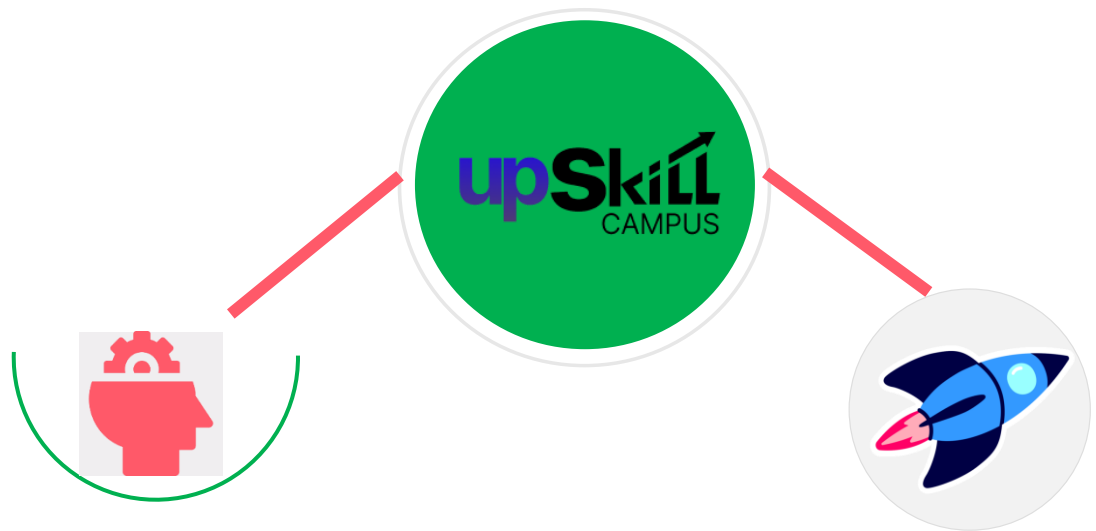
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

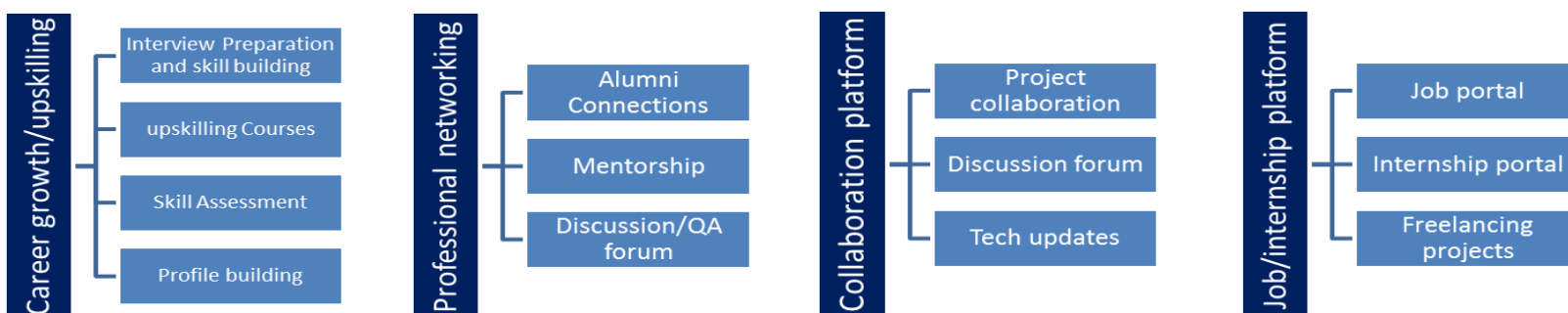
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.3 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.4 Reference

1. Python Documentation
2. Scikit-learn Documentation
3. Machine Learning Research Papers
4. Educational Data Mining Resources

2.5 Glossary

Terms	Acronym
AI	Artificial Intelligence
ML	Machine Learning
Dataset	Collection of data used for analysis
Linear Regression	Machine Learning algorithm used for prediction
Prediction	Estimating future outcomes using data

3 Problem Statement

Educational institutions often face difficulty in identifying students who may perform poorly in academics. Manual analysis of student performance consumes time and may not provide accurate predictions.

The objective of this project is to develop a Machine Learning model that predicts student academic performance based on study hours, attendance, assignments, and previous marks.

4 Existing and Proposed solution

Existing Solution

Traditional methods of student performance analysis are mostly manual and time-consuming. Teachers analyze marks and attendance records individually, which may not always provide accurate results.

Proposed Solution

The proposed solution uses Machine Learning techniques to automatically predict student performance. The Linear Regression algorithm is used to train the model using student data.

The system provides:

- Faster analysis
- Better prediction accuracy
- Early identification of weak students
- Data-driven decision making

4.1 Code submission (GitHub link) :

<https://github.com/Naveenjs5689/upskillCampus->

4.2 Report submission (GitHub link) :

<https://github.com/Naveenjs5689/upskillCampus->

5 Proposed Design/ Model

The proposed system is an AI-Based Student Performance Prediction System developed using Machine Learning techniques. The system analyzes student-related data such as study hours, attendance, assignments completed, and previous marks to predict the final academic performance of students.

The complete workflow of the system includes data collection, preprocessing, model training, prediction, and result evaluation.

5.1 High Level Diagram

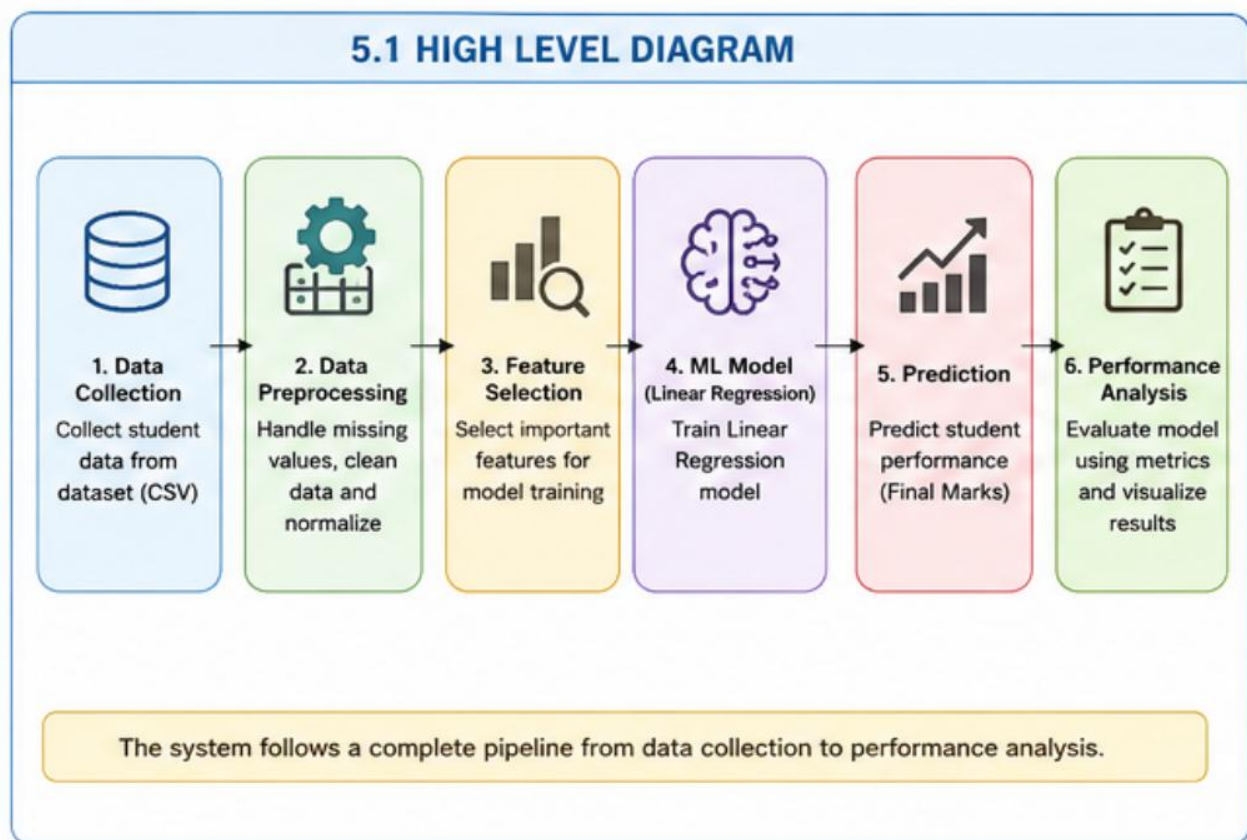
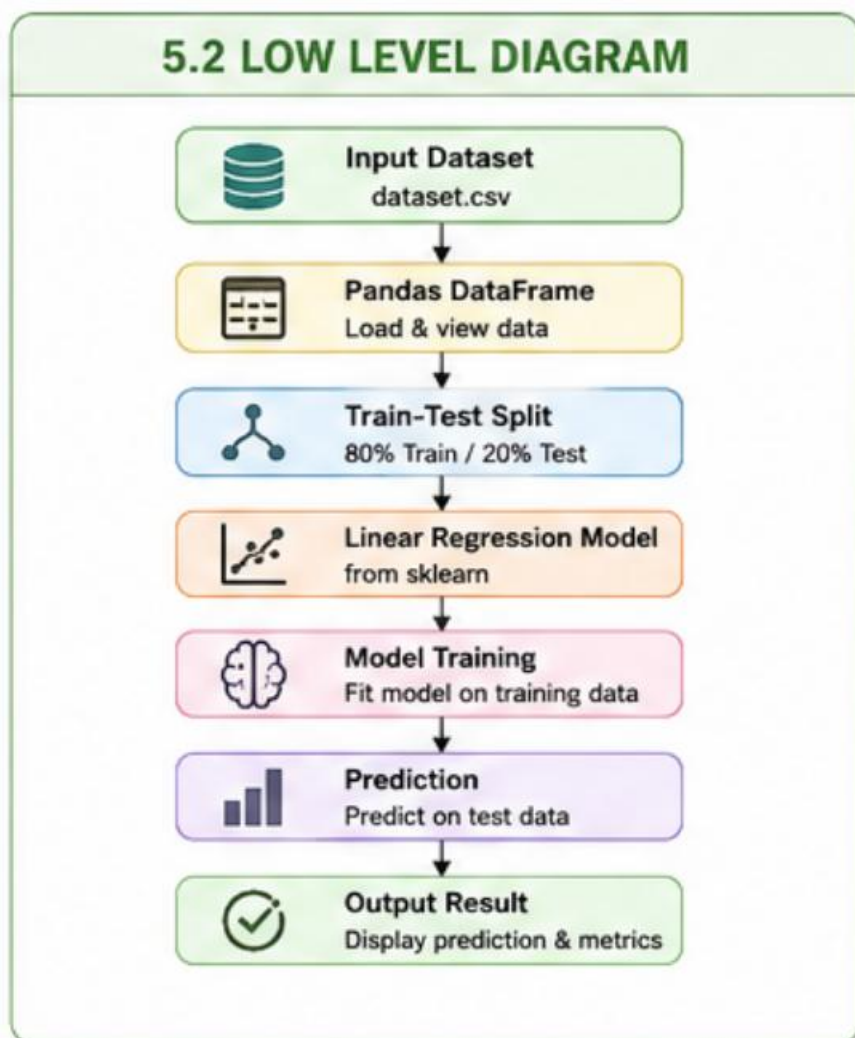
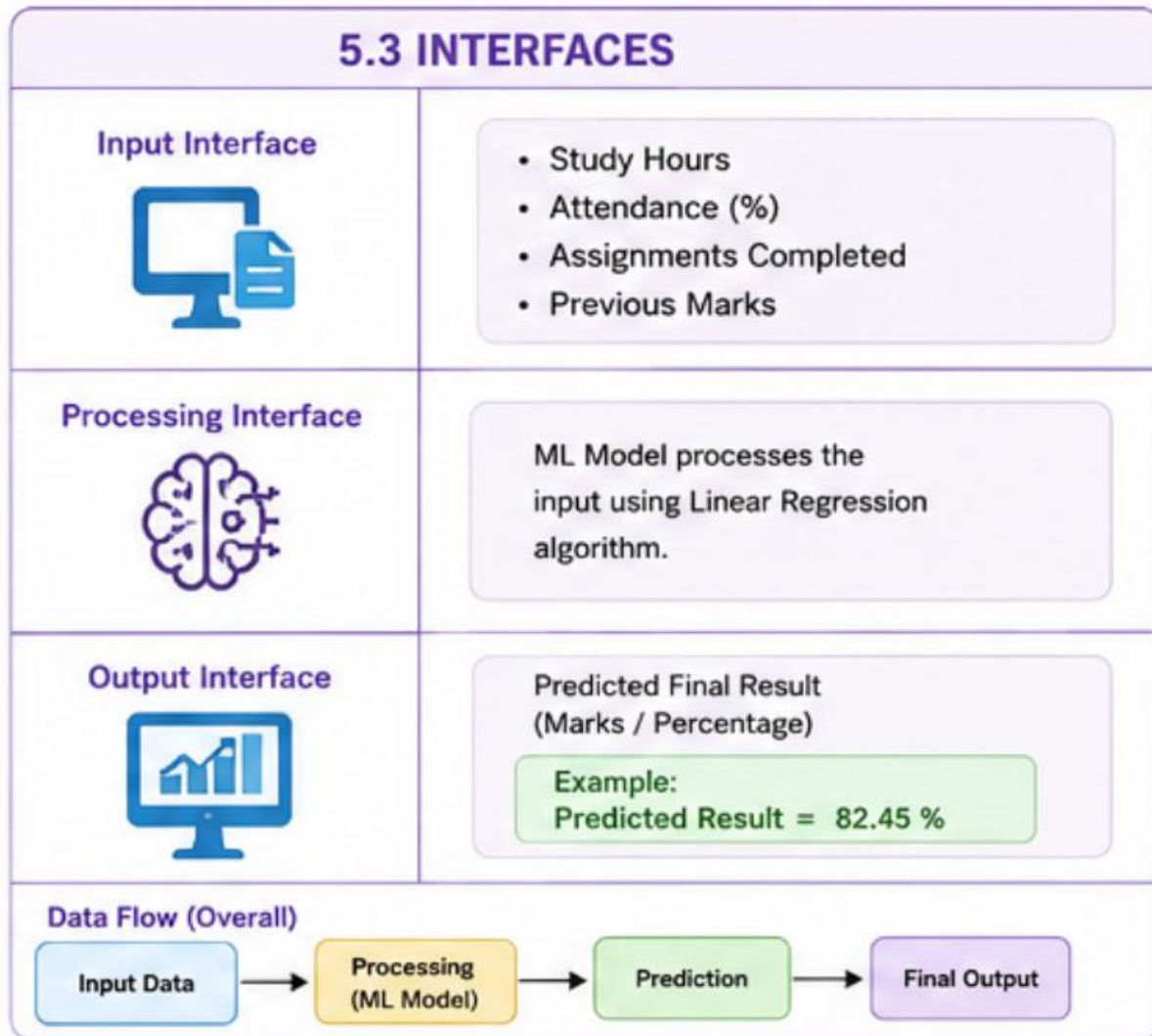


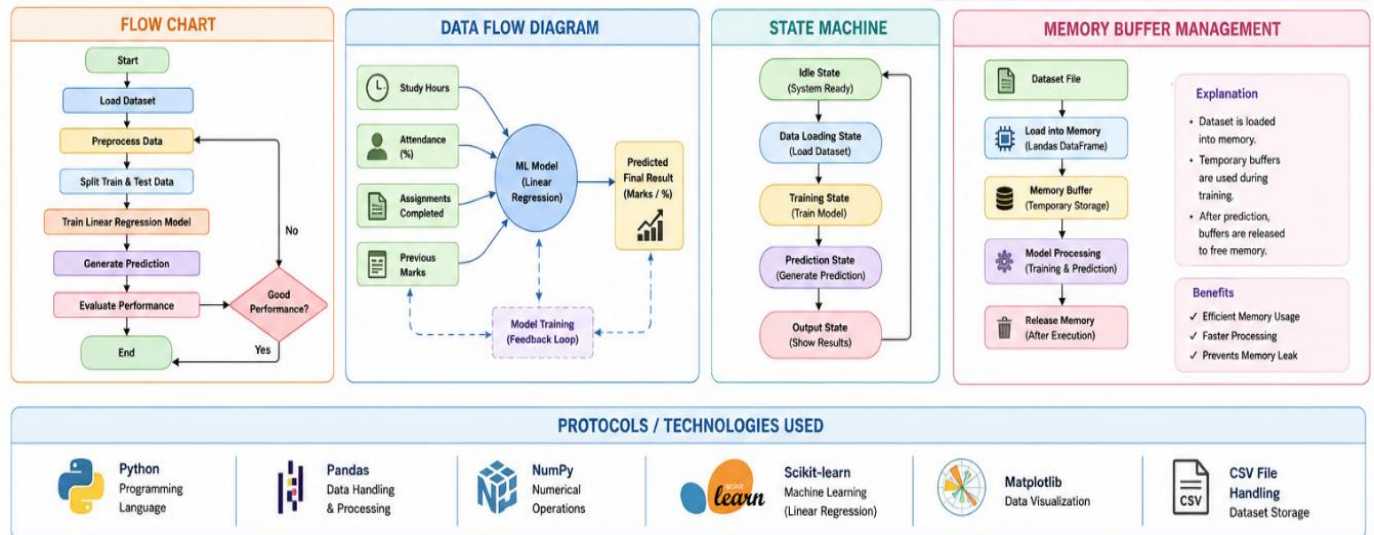
Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram



5.3 Interfaces





6 Performance Test

The performance testing phase evaluates the efficiency and accuracy of the AI-Based Student Performance Prediction System. The system was tested using student datasets to verify prediction accuracy, processing speed, and system reliability.

The main objective of performance testing is to ensure that the machine learning model produces accurate predictions with minimum processing time and memory usage.

6.1 Test Plan/ Test Cases

The performance testing phase evaluates the efficiency, accuracy, and reliability of the AI-Based Student Performance Prediction System.						
6.1 TEST PLAN / TEST CASES						
	Test Case ID	Test Scenario	Input	Expected Output	Actual Output	Status
	TC01	Dataset Loading	dataset.csv	Dataset should load successfully	Dataset loaded successfully	✓ PASSED
	TC02	Data Preprocessing	Student data	Missing values handled correctly	Data cleaned successfully	✓ PASSED
	TC03	Train ML Model	Training dataset	Model training successful	Model trained successfully	✓ PASSED
	TC04	Prediction Generation	Test dataset	Predicted results should display	Predictions displayed correctly	✓ PASSED
	TC05	Accuracy Evaluation	Predicted & actual values	Accuracy should be acceptable	Good prediction accuracy achieved	✓ PASSED
	TC06	Invalid Data Handling	Incorrect values	System should avoid crashes	System handled errors properly	✓ PASSED

6.2 Test Procedure

The following procedure was followed during testing:

1. The student dataset was loaded into the system.
2. Data preprocessing was performed using Pandas.
3. Training and testing datasets were created.
4. The Linear Regression model was trained using Scikit-learn.
5. Predictions were generated using test data.
6. The predicted results were compared with actual results.
7. System performance and accuracy were evaluated.

6.3 Performance Outcome

6.3 PERFORMANCE OUTCOME			
	Performance Parameter	Description	Result
	Algorithm Used	Machine learning algorithm used for prediction	Linear Regression
	Dataset Type	Type of dataset used	Student Academic Dataset
	Prediction Accuracy	Accuracy level achieved	Good
	Processing Speed	Time taken for prediction	Fast
	Memory Usage	Memory consumption during execution	Low
	System Stability	Stability during execution	Stable
	Error Handling	Ability to handle invalid inputs	Successful
	Scalability	Ability to handle larger datasets	Moderate
	Output Quality	Quality of prediction results	Accurate
	Overall Performance	Final system evaluation	Successful

7 My learnings

Internship 2026 provided valuable practical exposure to Artificial Intelligence, Machine Learning, and Data Science concepts. During this internship, I gained hands-on experience in developing a real-world project titled “AI-Based Student Performance Prediction System”.

Through this project, I learned how to collect, preprocess, analyze, and visualize data using Python libraries such as Pandas and NumPy. I also learned how to implement Machine Learning algorithms using the Scikit-learn library and generate predictions based on student academic data.

The internship improved my understanding of:

- Python programming
- Machine Learning concepts
- Data preprocessing techniques
- Model training and prediction
- Data visualization
- GitHub repository management
- Technical documentation and report preparation

I also learned the importance of teamwork, problem-solving, time management, and project planning while working on the internship tasks.

This internship helped me improve both my technical skills and professional skills, which will support my future career growth in the fields of Artificial Intelligence, Data Science, and Software Development.

Skills Learned

Technical Skills

Python Programming

Machine Learning

Data Analysis

Scikit-learn

GitHub Management

Report Documentation

Professional Skills

Communication Skills

Time Management

Teamwork

Problem Solving

Project Planning

Presentation Skills

Overall Experience

The internship experience was highly informative and beneficial. It provided practical industry exposure and helped me understand how Machine Learning can be applied to solve real-world educational problems. The guidance and support from upSkill Campus and UniConverge Technologies Pvt Ltd made this internship a valuable learning experience.

8 Future work scope

The AI-Based Student Performance Prediction System can be further improved by adding advanced features and technologies in future versions. Although the current system successfully predicts student academic performance using Machine Learning techniques, there are several opportunities for enhancement.

Future improvements can increase prediction accuracy, system efficiency, and real-time usability.

-
- **Future Enhancements**
 - **1. Advanced Machine Learning Algorithms**

Future versions can implement advanced algorithms such as:

- Random Forest
- Decision Tree
- Support Vector Machine (SVM)
- Deep Learning Models

These algorithms may improve prediction accuracy for larger datasets.

-
- **2. Web-Based Application**

The system can be converted into a web application using:

- Flask
- Django
- Streamlit

This will allow teachers and students to access predictions online easily.

- **3. Real-Time Student Monitoring**

Future systems can monitor student performance in real time by integrating:

- Attendance systems
- Online test scores
- Assignment portals

This helps institutions track student progress continuously.

- **4. Larger and Real Datasets**

Using larger datasets collected from schools and colleges can improve:

- Model training
 - Prediction reliability
 - Overall system performance
-

- **5. Data Visualization Dashboard**

Interactive dashboards can be added using:

- Power BI
- Tableau
- Matplotlib dashboards

This helps teachers visualize student performance effectively.

- **6. Mobile Application Support**

A mobile application can be developed for Android and iOS platforms to provide:

- Easy access to predictions
- Student progress tracking
- Instant notifications

- **7. Cloud Integration**

The project can be deployed on cloud platforms such as:

- AWS
- Microsoft Azure
- Google Cloud

Cloud deployment improves scalability and accessibility.

- **Expected Future Benefits**

Enhancement	Benefit
Advanced ML Models	Higher prediction accuracy
Web Application	Easy online access
Real-Time Monitoring	Continuous student tracking
Large Datasets	Better model performance
Dashboards	Improved data visualization
Mobile App	Easy accessibility
Cloud Deployment	High scalability

- **Conclusion of Future Scope**

The future scope of this project is very wide in the education sector. By integrating advanced Machine Learning techniques, cloud technologies, and real-time monitoring systems, the project can become a complete smart educational analytics solution for schools and colleges.